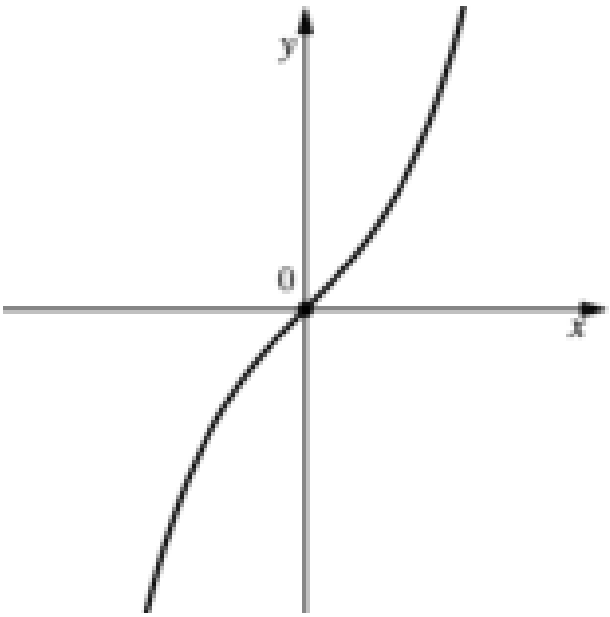
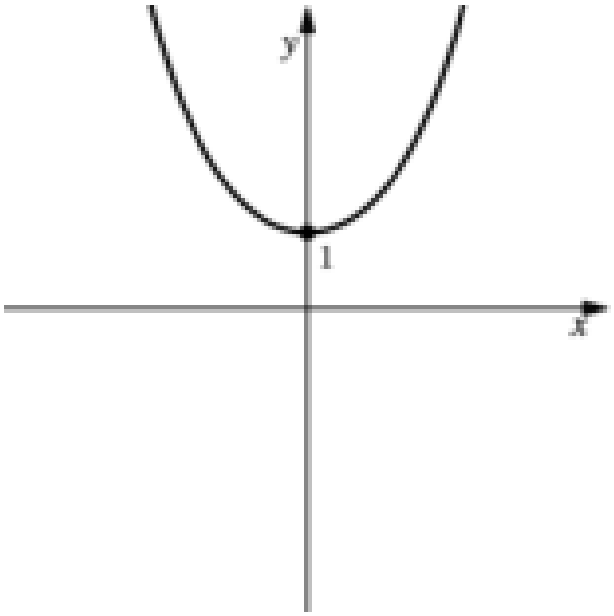
1.31 Graph Sketching: Hyperbolic Functions

Worked Solutions

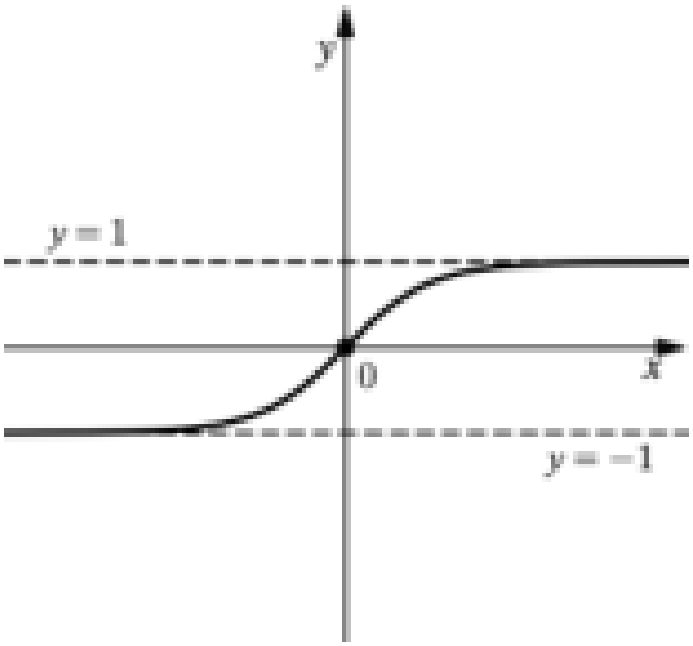
1 a)



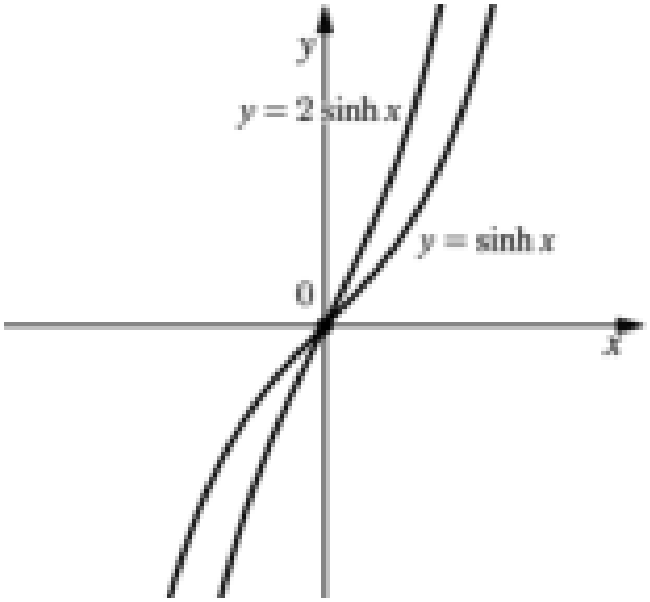
1 b)



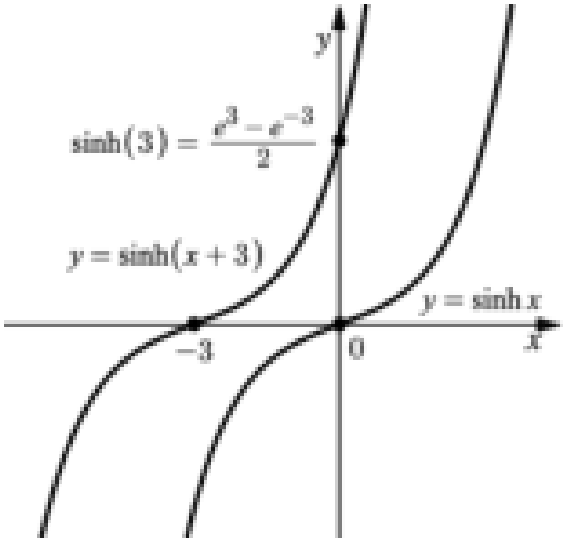
1 c)



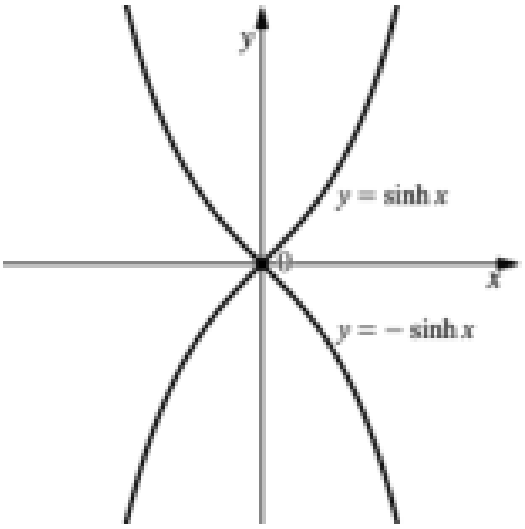
2 a)



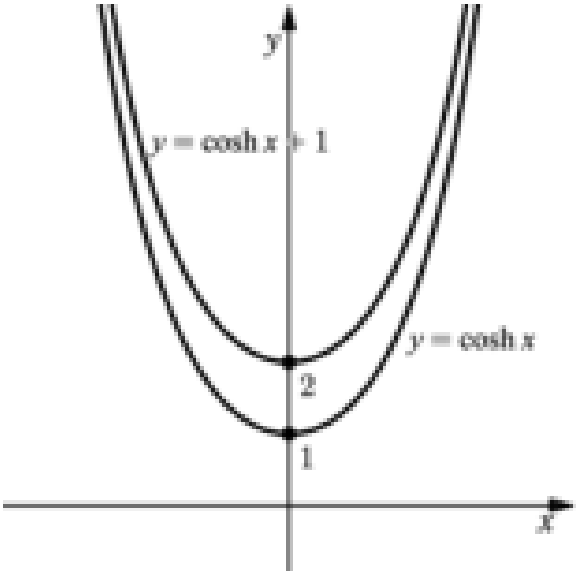
2 b)



2 c)

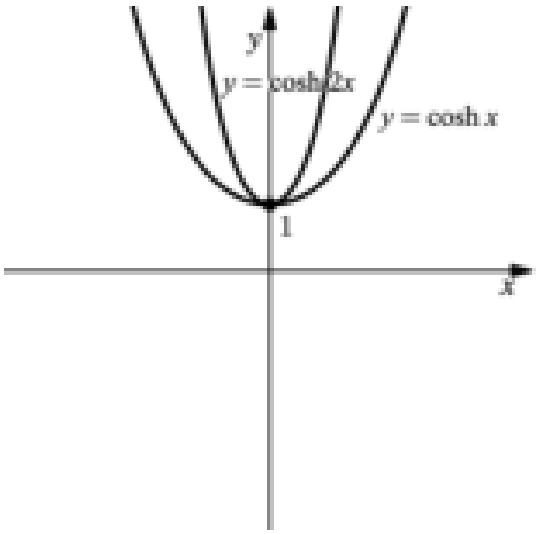


2 d)

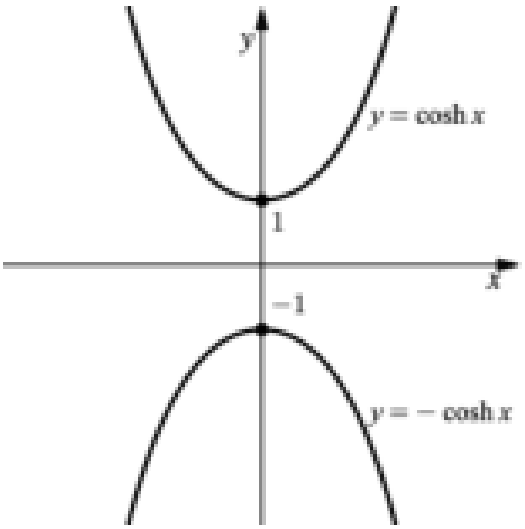


2 e)

1



2 f)



3 a)

$y = \sinh x + b$

Substitute $(\ln(\sqrt{2}-1), 0)$:

$$0 = \sinh(\ln(\sqrt{2}-1)) + b$$

$$\Rightarrow 0 = \frac{e^{\ln(\sqrt{2}-1)} - e^{-\ln(\sqrt{2}-1)}}{2} + b$$

$$\Rightarrow 0 = e^{\ln(\sqrt{2}-1)} - e^{\ln((\sqrt{2}-1)^{-1})} + 2b$$

$$\Rightarrow 0 = \sqrt{2} - 1 - \frac{1}{\sqrt{2}-1} + 2b$$

$$\Rightarrow 0 = \sqrt{2} - 1 - (1+\sqrt{2}) + 2b \Rightarrow 0 = \sqrt{2} - 1 - 1 - \sqrt{2} + 2b$$

$$\Rightarrow 2b = 2 \Rightarrow b = 1$$

$y = \sinh x + 1$

3 b)

$y = \sinh(x - a)$

Substitute $(0, \frac{1-e^4}{2e^2})$:

$$\frac{1-e^4}{2e^2} = \sinh(-a) \Rightarrow \frac{1-e^4}{2e^2} = \frac{e^{-a} - e^{-(-a)}}{2}$$

$$\Rightarrow \frac{1-e^4}{e^2} = e^{-a} - e^{a} \Rightarrow \frac{1}{e^2} - \frac{e^4}{e^2} = e^{-a} - e^{a}$$

$$\Rightarrow e^{-2} - e^{2} = e^{-a} - e^{a} \Rightarrow a = 2$$

$y = \sinh(x - 2)$

4 a)

$y = \cosh x + b$

Substitute $(\ln(3+2\sqrt{2}), 0)$:

$$0 = \cosh(\ln(3+2\sqrt{2})) + b$$

$$\Rightarrow 0 = e^{\ln(3+2\sqrt{2})} + e^{-\ln(3+2\sqrt{2})} + 2b$$

$$\Rightarrow 0 = e^{\ln(3+2\sqrt{2})} + e^{\ln((3+2\sqrt{2})^{-1})} + 2b$$

$$\Rightarrow 0 = 3 + 2\sqrt{2} + \frac{1}{3+2\sqrt{2}} + 2b$$

$$\Rightarrow 0 = 3 + 2\sqrt{2} + 3 - 2\sqrt{2} + 2b \Rightarrow 0 = 6 + 2b \Rightarrow b = -3$$

$\therefore y = \cosh x - 3$

4 b)

$y = \cosh(x - a)$

Substitute $(0, \frac{5}{4})$:

$$\frac{5}{4} = \cosh(-a) \Rightarrow \frac{5}{4} = \frac{e^{-a} + e^{-(-a)}}{2} \Rightarrow \frac{5}{2} = e^{-a} + e^{a}$$

$$\Rightarrow \frac{5}{2}e^{a} = 1 + e^{2a} \Rightarrow 0 = e^{2a} - \frac{5}{2}e^{a} + 1$$

$$\Rightarrow 0 = 2e^{2a} - 5e^{a} + 2 \Rightarrow 0 = (2e^{a} - 1)(e^{a} - 2)$$

$$\Rightarrow e^{a} = \frac{1}{2} \text{ or } e^{a} = 2 \Rightarrow a = \ln\frac{1}{2} \text{ or } a = \ln 2 \Rightarrow a = \pm \ln 2$$

$\therefore y = \cosh(x + \ln 2)$

5 a)

$y = \tanh x$ has horizontal asymptotes at $y = -1$ and $y = 1$.

$y = \tanh x \mapsto y = \tanh(x-2) + 5$ is a translation by the vector $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$, so the horizontal asymptotes are translated 5 units upwards.

$\therefore y = \tanh(x-2) + 5$ has horizontal asymptotes at $y = 4$ and $y = 6$.

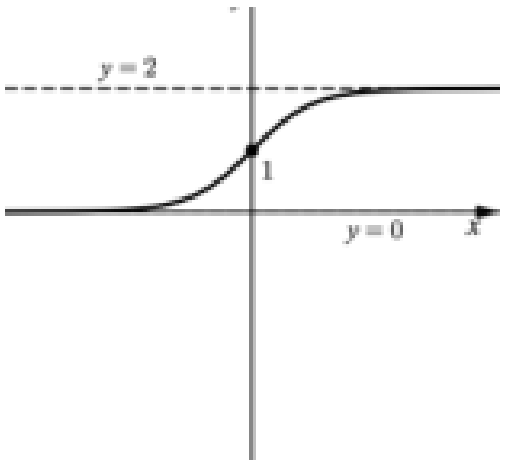
5 b)

$y = \tanh x$ has horizontal asymptotes at $y = -1$ and $y = 1$.

$y = \tanh x \mapsto y = \tanh(x+\frac{5}{2}) - \frac{3}{2}$ is a translation by the vector $\begin{pmatrix} -\frac{5}{2} \\ -\frac{3}{2} \end{pmatrix}$, so the horizontal asymptotes are translated $\frac{3}{2}$ units downwards.

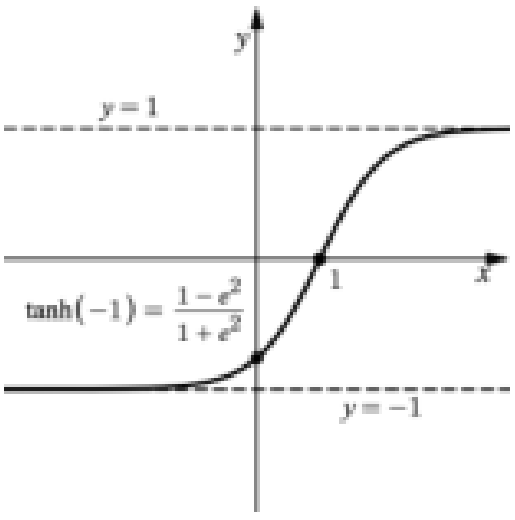
$\therefore y = \tanh(x + \frac{5}{2}) - \frac{3}{2}$ has horizontal asymptotes at $y = -\frac{5}{2}$ and $y = -\frac{1}{2}$.

6 a)

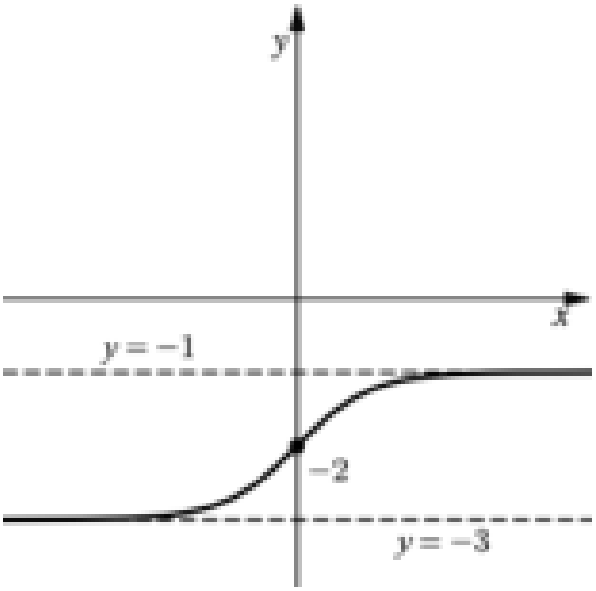


6 b)

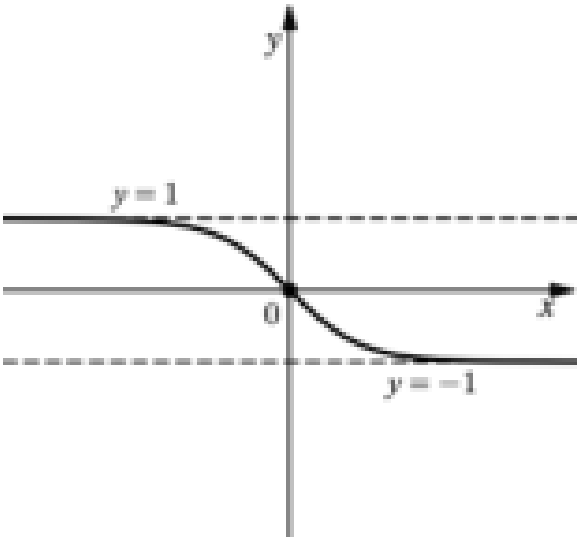
:ching: Hyperbolic Functions (Worked S



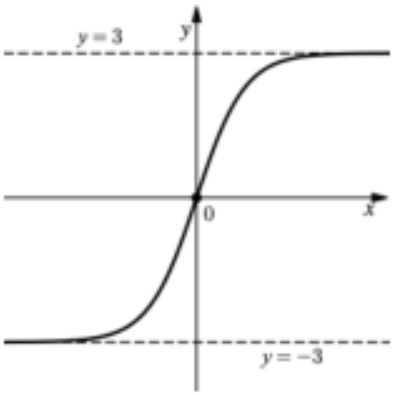
6 c)



6 d)

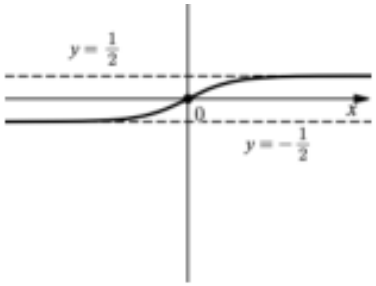


7 a)





7 b)



8 a)

$$\frac{3+1}{2} = 2$$

$y = \tanh x + 2$

8 b)

$$\frac{\frac{1}{2}+(-\frac{3}{2})}{2}=-\frac{1}{2}$$

$y=\tanh x-\frac{1}{2}$

8 c)

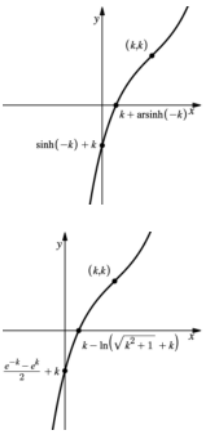
$y=\tanh(x-a)$

Substitute $(0,\frac{e^4-1}{e^4+1})$:

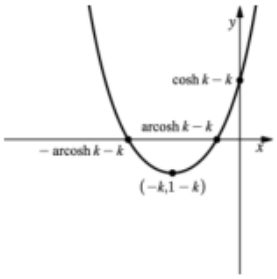
$$\frac{e^4-1}{e^4+1}=\tanh(-a)\Rightarrow\frac{e^4-1}{e^4+1}=\frac{e^{-2a}-1}{e^{-2a}+1}$$
$$\Rightarrow -2a=4\Rightarrow a=-2$$

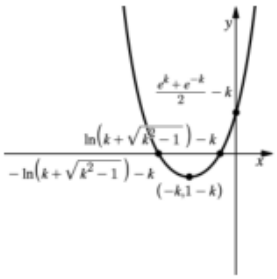
$\therefore y=\tanh(x+2)$

9 a)

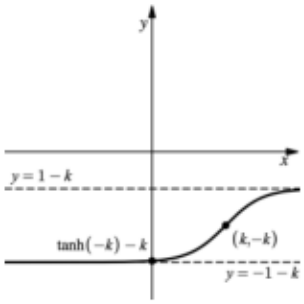


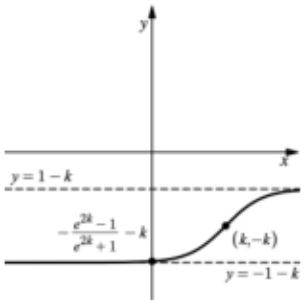
9 b)


y
cosh k − k
arcosh k − k
− arcosh k − k
x
(−k,1 − k)

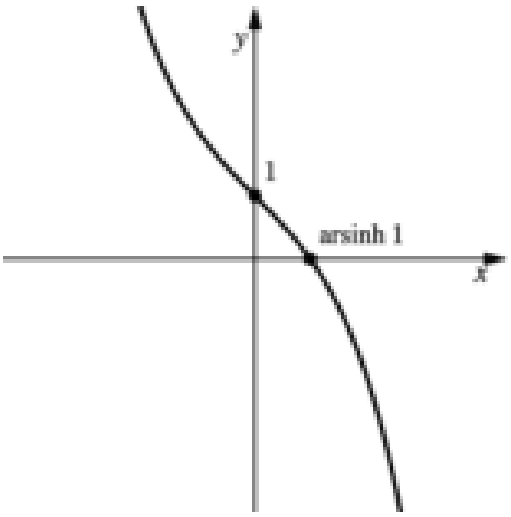

y
$\frac{e^k + e^{-k}}{2} - k$
$\ln\left(k + \sqrt{k^2 - 1}\right) - k$
x
$-\ln\left(k + \sqrt{k^2 - 1}\right) - k$
(−k,1 − k)

9 c)

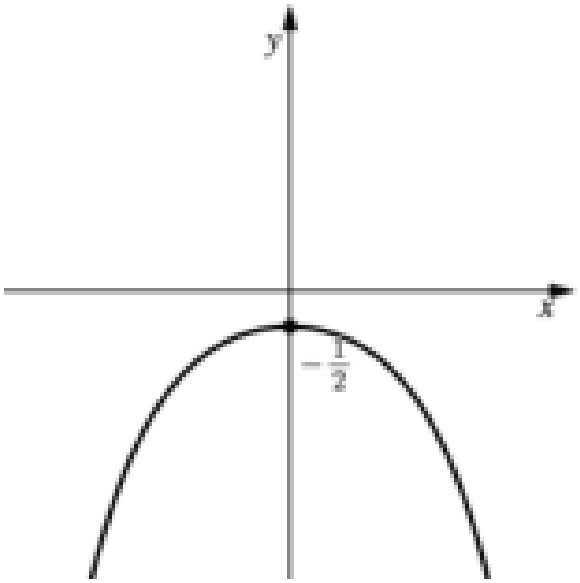

y
x
y = 1 − k
tanh(−k) − k
(k,−k)
y = −1 − k


y
x
y = 1 − k
$-\frac{e^{2k} - 1}{e^{2k} + 1} - k$
(k,−k)
y = −1 − k

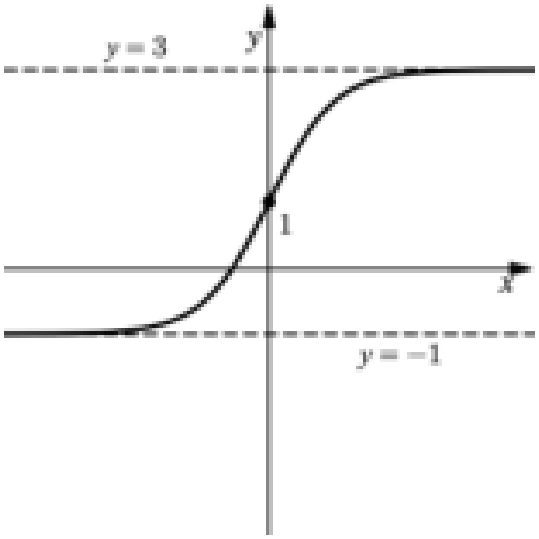
10 a)



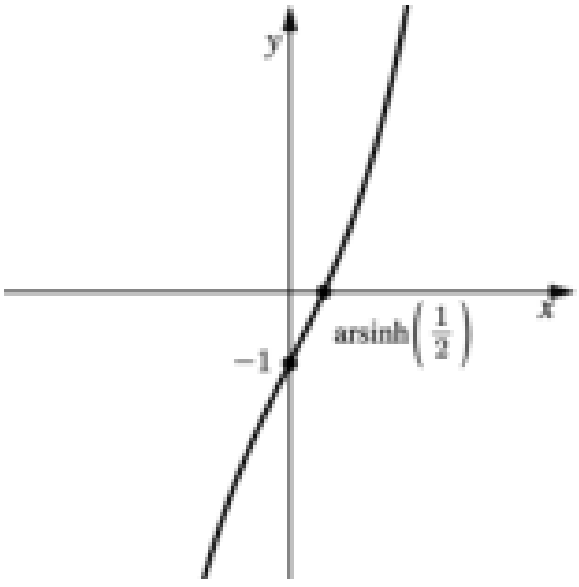
10 b)



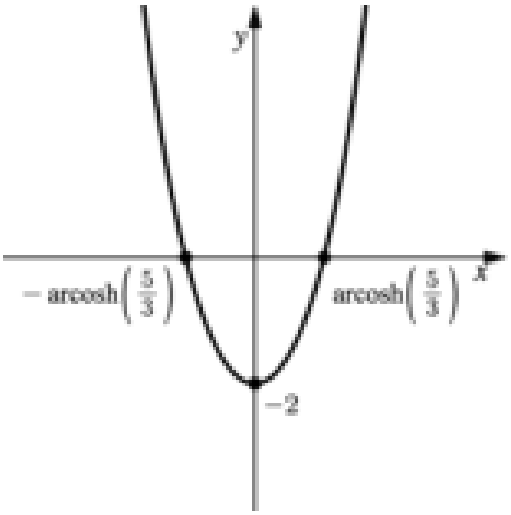
10 c)



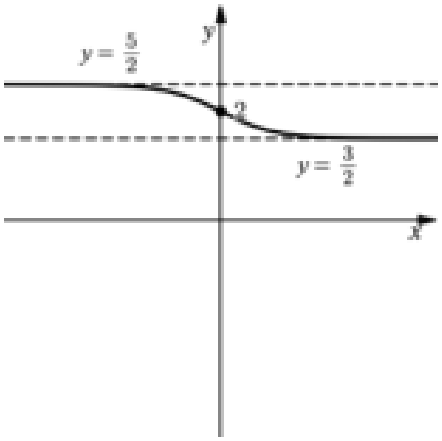
10 d)



10 e)



10 f)



11 a)

First consider $y = \cosh x$:

Reflect in the x-axis to get $y = -\cosh x$.

Translate by $\begin{pmatrix} 0 \\ k \end{pmatrix}$ to get $y = k - \cosh x$.

Translate by $\begin{pmatrix} -k \\ 0 \end{pmatrix}$ to get $y = k - \cosh(x + k)$.

Check the curve pass through $(0, k - \frac{e^k+e^{-k}}{2})$:

$$k - \cosh(0 + k) = k - \cosh k = k - \frac{e^k + e^{-k}}{2} \text{ as required.}$$

Alternative Method:

Using the y-intercept: $k - \frac{e^k+e^{-k}}{2} \equiv k - \cosh k$.

This must be the value when $x = 0$, hence: $y = k - \cosh(x + k)$.

Check the curve pass through $(-k, k - 1)$:

$$k - \cosh(-k + k) = k - \cosh 0 = k - 1 \text{ as required.}$$

11 b)

First consider $y = \cosh x$:

Reflect in the x-axis to get $y = -\cosh x$.

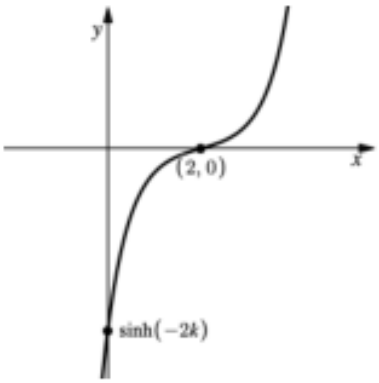
Stretch, parallel to the y-axis, scale factor k to get $y = -k \cosh x$.

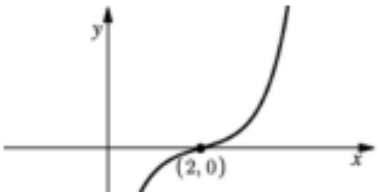
Translate by $\begin{pmatrix} 0 \\ 2k \end{pmatrix}$ to get $y = 2k - k \cosh x$.

Check the curve passes through $(\ln(2 + \sqrt{3}), 0)$:

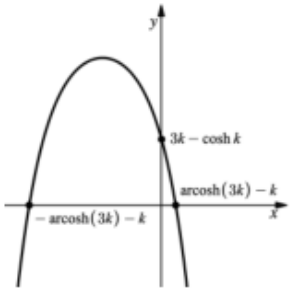
$$\begin{aligned}
&2k - k\cosh(\ln(2 + \sqrt{3})) \\
&= 2k - k \times \frac{e^{\ln(2+\sqrt{3})} + e^{-\ln(2+\sqrt{3})}}{2} \\
&= 2k - k \times \frac{e^{\ln(2+\sqrt{3})} + e^{\ln((2+\sqrt{3})^{-1})}}{2} \\
&= 2k - \frac{1}{2}k(e^{\ln(2+\sqrt{3})} + e^{\ln((2+\sqrt{3})^{-1})}) \\
&= 2k - \frac{1}{2}k(2 + \sqrt{3} + \frac{1}{2+\sqrt{3}}) \\
&= 2k - \frac{1}{2}k(2 + \sqrt{3} + \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}) \\
&= 2k - \frac{1}{2}k(2 + \sqrt{3} + \frac{2-\sqrt{3}}{4-3}) \\
&= 2k - \frac{1}{2}k(2 + \sqrt{3} + 2 - \sqrt{3}) = 2k - \frac{1}{2}k \times 4 = 0 \text{ as required.}
\end{aligned}$$

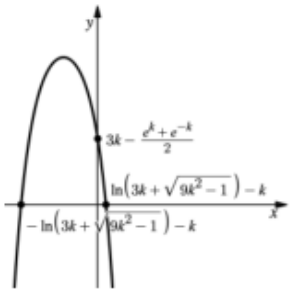
12 a)


y
x
(2, 0)
sinh(−2k)

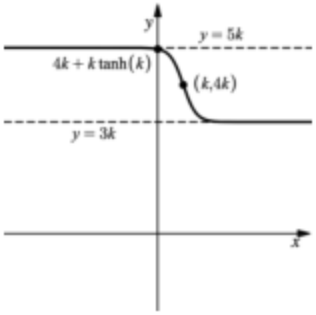

y
x
(2, 0)

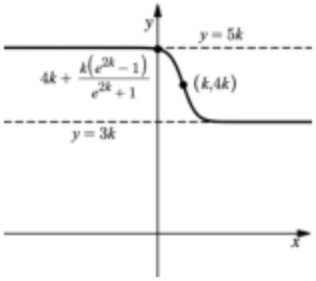
12 b)


y
3k − cosh k
arcosh(3k) − k
x
− arcosh(3k) − k


y
3k − (e^k + e^−k)/2
ln(3k + √(9k^2 − 1)) − k
x
− ln(3k + √(9k^2 − 1)) − k

12 c)


y
y = 5k
4k + k tanh(k)
(k,4k)
y = 3k
x


y
y = 5k
4k + k(e^{2k} − 1)/(e^{2k} + 1)
(k,4k)
y = 3k
x